Full Name
Section

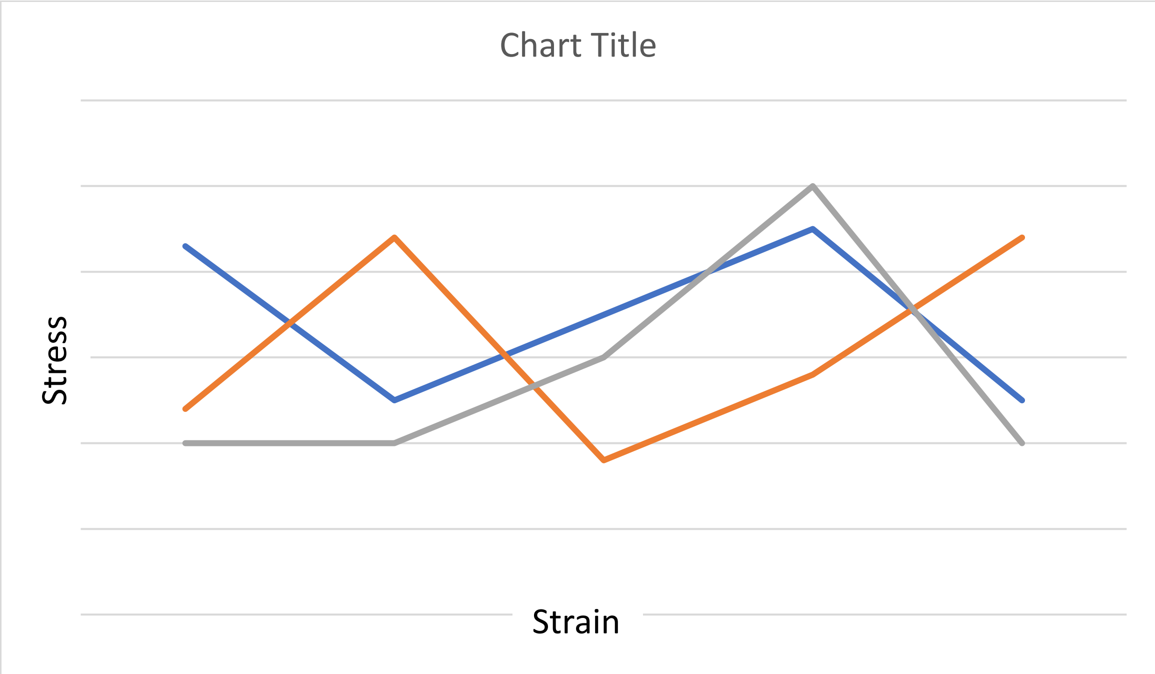
Tensile Test Report Template

In all questions: You can use Excel, Matlab, Python, or anything else you are comfortable with. Your report should be approximately this long; therefore, be concise in your explanations.

Specimen 1: Metal

(a) All groups' data for tensile test experiment will be provided on Moodle. Using the provided data files, plot engineering stress-strain curves in the same figure. Comment on possible reasons behind observed differences. Indicate necessary units. (The figure is just an example to show your figure's approximate size. Graphs will overlap; so individual data points do not need to be readable)

Remark: Beware that excel data gives you %elongation, meaning: $\frac{\Delta l}{l_0} \times 100$

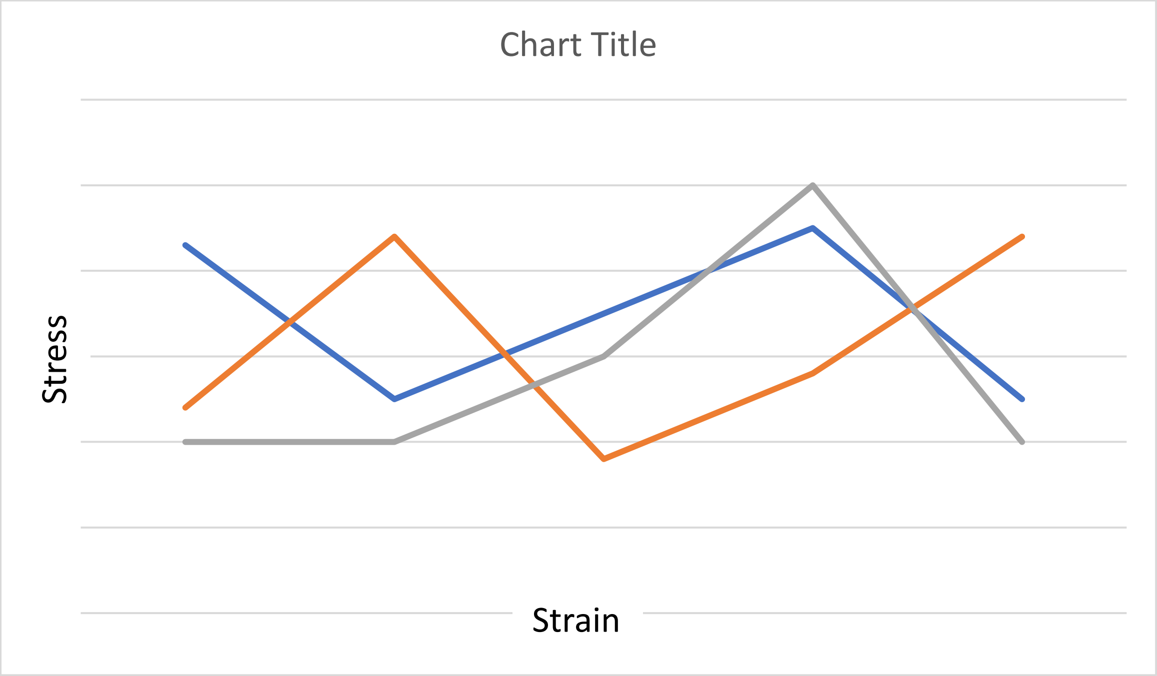


(b) Using the data, calculate mean values for: Young's Modulus (E), 0.2% offset yield strength (σ_0), and ultimate tensile strength (σ_u). Explain the procedure you conduct in a few sentences:

- How do you process data to find Young's modulus?
- How do you process data to find 0.2% offset yield strength (σ_0)?
- How do you process data to find ultimate tensile strength (σ_u)?
- Compare the mean values with the results you found from your group's data(identify which day&group). You can use the values in Excel "Results" sheet to check if your calculations make sense but use your own calculations for comparison. (In the Excel Results sheet E: Young's modulus, Rp0.2: 0.2% offset yield strength, Rm: ultimate tensile strength)

(c) Comment on the dependence of the yield strength on the amount of offset. In particular, compare the yield strength values for offset values of 0.1%, 0.2% and 0.3%. Use your group's data.

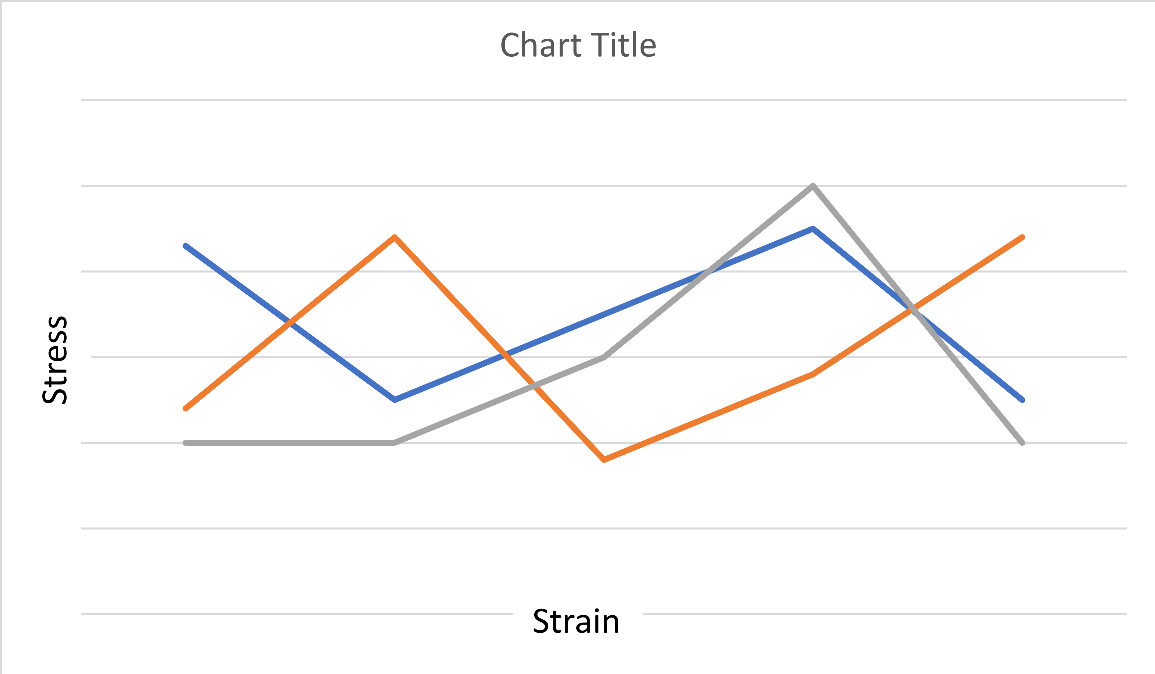
(d) Choose one of the engineering stress-strain curves from part (a) and calculate the corresponding true stress-strain curve. Plot both curves on the same figure. Indicate necessary units. (The figure is just an example to show your figure's approximate size. Individual data points must be readable)



(e) Search online about Lüders bands and explain why this concept is relevant to the test results.

Specimen 2: Polymer

(a) All groups' data for tensile test experiment is provided on Moodle. Using the provided data files, plot engineering stress-strain curves in the same figure. (The figure is just an example to show your figure's approximate size. Graphs will overlap; so individual data points do not need to be readable)



(b) Using the provided data, calculate a mean value for ultimate tensile strength (σ_u). Compare the mean values with the results you found from your group's data (identify which day&group). You can use the values in Excel "Results" sheet to check if your calculations make sense but use your own calculations for comparison. (In the Excel Results sheet σ_m: ultimate tensile strength)

Remark: About the polymer data: it seems like excel sheets already cut the peaks off, so you can use the whole data in the excel documents (discard the previous email about the peaks).

(c) Give a physics-based explanation of the observed differences between the stress-strain curves for the metal and polymer specimens.

(d) Comment on the qualitative and quantitative dependence of the polymer curve on the strain-rate.

(e) How does the necking behavior of polymer differ from the necking behavior observed for steel?

References

Appendix

Include your code & extra figures